

Java Backend Rapid-Fire — 5 Core Questions

Q1 — Who creates the OS process?

- Terminal calls OS (fork + exec).
 - OS loads the java binary.
 - JVM runs inside that process.
 - Mental model: Terminal → OS → JVM.
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Q2 — Why multiple threads before main()?

- JVM initializes runtime.
 - Creates GC threads.
 - Creates JIT/compiler threads.
 - Creates main thread before executing user code.
 - Mental model: JVM runtime starts before your code.
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Q3 — Many BLOCKED threads: first move?

- Find repeated 'waiting to lock <0x...>'.
 - Search for 'locked <0x...>' (lock owner).
 - Inspect lock owner's stack trace.
 - Mental model: Find monitor → Find owner → Inspect code.
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Q4 — What exhausts first in Spring MVC + Tomcat?

- Request threads (http-nio-*).
 - Usually due to slow DB or external calls.
 - Mental model: Request threads are the scarce resource.
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Q5 — Why is @Async + .get() bad?

- Blocks request thread.
 - Defeats async.
 - Can cause thread pool starvation.
 - Mental model: Async + block = fake async.
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